

Bounded and Reproducible Progression Synthesis for Brain Metastases: A Lightweight Two-Stage Latent Diffusion Model with Generation-Isolation Audit

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Code & checkpoints: github.com/jgn2pyh2wn-afk/brain_tumour_predict

1. Motivation

Brain metastases are the commonest intracranial secondary tumours, and management turns on how lesions change between follow-ups, yet repeat imaging is scarce. We contribute a generation-isolation audit (§7), which re-runs every reported number under surrogates to test whether it really comes from the sampler, and a lightweight two-stage latent diffusion model (LDM) [2] that synthesises T1-Gd progression as a bounded, reproducible 64^3 sequence from one scan; its step axis is a generation grid, not clinical time. Small by design — a custom-trained 3D VAE and DiT [3], not a biophysically conditioned framework [4].

2. Method

A 3D convolutional autoencoder maps a $64 \times 64 \times 64$ T1-Gd volume to a $4 \times 16 \times 16 \times 16$ latent; a Diffusion Transformer (dim 64, 2 blocks, 4 heads, patch $2 \times 2 \times 2$, 512 tokens; $T = 200$, linear $\beta \ 1 \times 10^{-4} \rightarrow 0.02$ [1]) predicts latent noise conditioned on a globally pooled latent. Stage 1 picks the starting noise level from the prescribed step span (AdaNI); stage 2 re-noises to 0.157 and refines texture. A structure-preserving post-process confines the residual to a lesion-candidate mask and clips it at $c = 0.08$ with strength $s = 0.35$; all components are custom-trained.

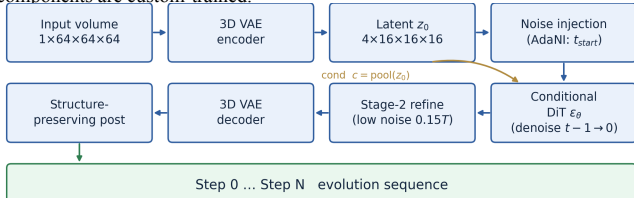


Fig. 1. Two-stage latent-diffusion framework (schematic); every component is custom-trained.

3. Growth-law time weighting

Endpoint weights follow an explicit growth law (exponential / Gompertz / logistic-Fisher-KPP) with an equivalent doubling time T_d instead of uniform step/ N interpolation; this is a monotone re-parameterisation of the step axis, not a cohort-calibrated growth model, but one with closed-form structure. Five derived properties are verified on all 51 cases — load-weight affinity (residual 1.02×10^{-15}), a convex-combination identity (1.92×10^{-7}), an adjacent-increment bound, the closed-form maximiser ($r^* = 56.30$, rel. 4.4×10^{-8}) and boundedness and $\|x(t) - x_0\| = w(t) \cdot \|x_N - x_0\|$ with $|\Delta_{\text{gen}}| \leq s \cdot c = 0.028$.

4. Results: structure preservation (Fig. 2)

As trained (before the §8 fix), on $N = 100$ steps over 51 cases: soft contour IoU 0.9836 $\pm$ 0.0047, hard IoU 0.9881 $\pm$ 0.0035, texture correlation 0.9808 $\pm$ 0.0012, **candidate-region** mean change -0.0219 ± 0.0023 — the region is the heuristic candidate field (24.4% of brain), **not** the lesion; outside/inside ratio 0.633 (0.416–0.863); pre-clip reconstruction PSNR 15.993 $\pm$ 1.497 dB.

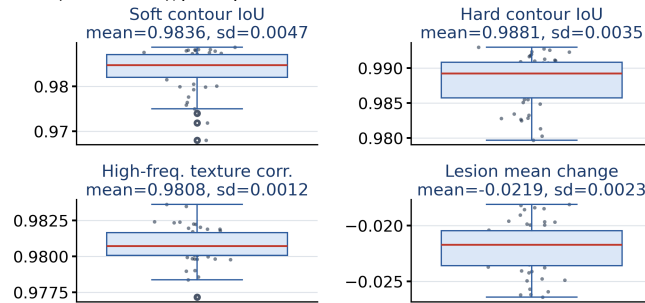


Fig. 2. Progression metrics over the 51-case test set (as-trained weights, i.e. before the §8 fix; $N=100$ steps; box=IQR, dots=per-case).

5. Expert-lesion cross-check (n = 105; not held-out)

Expert *seg* labels exist only in the training split (the 105 studies it trained on): a cross-check, not held-out performance. The candidate field is $125 \times$ the lesion, covering 29.0% of lesion voxels though only 0.37% of it is lesion. Inside expert masks (trilinear / max-pool) the same mean change is $-0.0494 \pm 0.0237 / -0.0239 \pm 0.0090$ vs. -0.0214 ± 0.0025 : up to $2.3 \times$, and the §7 audit does not move.

6. Qualitative progression (Fig. 3)

Three cases, Step 0 $\rightarrow$ 100, through the code's own display chain (per-slice percentile stretch; stored data intact). Black marks voxels where $m \times w(t) > 0.35$,

m the heuristic lesion-candidate field: a **prescribed overlay, not predicted growth**, on the axial slice with most mask support ($Z = 26/27/38$).

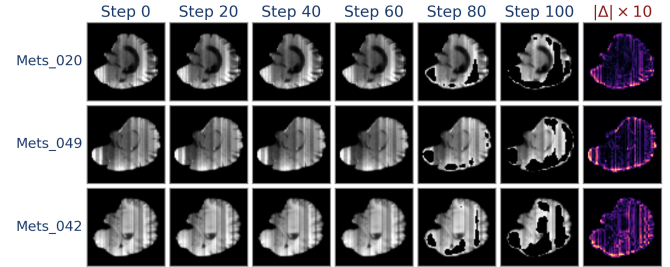


Fig. 3. Three cases, Step 0 $\rightarrow$ 100. Black = the **prescribed mask overlay, not predicted growth**. Right: $10 \times$ the model's own change (raw intensity; saturates at 0.1).

7. Generation-isolation audit (Fig. 4)

As trained, replacing the sampler with frozen sampling, a constant latent or another patient's result leaves **all four metrics bit-identical**, while source copy and pure noise do (our trivial baselines). This survives the region fix of §5: on the 105 labelled studies all five sampler-side surrogates give **identical** mean change (-0.0494 ± 0.0237), and deleting generation moves it by 16%. Root cause: the decoder's Sigmoid saturates (latent magnitude 11.43; 99.97% of candidate-core voxels exceed the 0.08 clip), so §4–§5 track the post-processing chain, not the sampler.

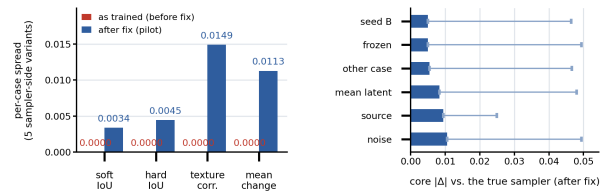


Fig. 4. Sampler attribution before (a) and after (b) the §8 fix.

8. Verified fix: sampler attribution restored

Two changes only (EMA latent standardisation; Hardtanh(0,1) decoder output) under an otherwise identical recipe (4.5 min pilot) give reconstruction PSNR 15.993 $\pm$ 1.497 $\rightarrow$ **29.299 $\pm$ 1.318 dB** and decoder std 0.0010 $\rightarrow$ 0.1447. Re-running the audit on all 51 cases, the five sampler-side variants stop being bit-identical (Fig. 4): the sampler now moves every reported metric, with a post-processed core difference from the true sampler of 0.0050–0.0106. The generation is still not lesion-specific (core/ring 0.94).

9. Take-home and limits

Contribution. A generation-isolation audit that reports which numbers survive sampler replacement; bounded, reproducible progression synthesis by a lightweight two-stage LDM, with released code, weights, scripts and checker; expert-lesion calibration of a heuristic mask; and a two-change decoder fix that restores sampler attribution on all 51 cases. **Limits.** No longitudinal ground truth (the step index is a generation condition, not clinical time); T_d uncalibrated; volumes resampled to 64^3 , so spacing is not modelled; the expert-lesion check uses training studies, as labels exist only there.

Compliance with Ethical Standards

Data and ethics. BrainMetShare [5] (156 studies; publisher's split 105/51), non-commercial research licence. Retrospective, de-identified secondary analysis: no new data, no ethics approval or consent required. **Conflicts of interest / funding:** none. **Reproducibility.** Code, weights and scripts released; a paper-to-code checker re-derives every claim (246 pass / 0 fail). **AI disclosure.** OpenAI Codex was used for (i) *code assistance* (training, inference, audit and figure scripts) and (ii) *paper layout and typesetting*; every number is emitted by the released code, and the author directed and edited all content.

References

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